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UNIVERSITI MALAYSIA SARAWAK

Faculty of Computer Science and Information Technology

Assignment/Report Cover Sheet

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1. Methodology

1.1 Simulation Environment

All simulations were run using NS-3 version 3.41 on an Ubuntu 22.04 LTS machine. NS-3 stands for Network Simulator 3, which is a discrete event simulator that is maintained by the ns-3 consortium. Each simulation case was programmed as a separate .cc file in the scratch/ directory.

1.2 Performance Metrics

Metric	Definition	Used In
Throughput (Mbps / Kbps)	Total received bytes × 8 / flow duration	Tasks 1, 2
Packet Delivery Ratio (PDR)	(Rx packets / Tx packets) × 100%	Task 2
Latency / Channel Delay	One-way propagation delay configured on the channel	Task 1
Bit / Packet Error Rate	Probability of packet drop applied by RateErrorModel	Task 1
Congestion Window (cWnd)	Sender's TCP cWnd in bytes, traced over time	Task 3

1.3 Workload Division

Team Member	Responsibility
Aaron	Designed and ran the Point-to-Point experiments (Task 1).
Alif	Implemented the CSMA contention study and NetAnim trace (Task 2).
Goh	Built the dumbbell topology and traced NewReno vs CUBIC cWnd (Task 3).
Azim	Integrated results, drafted the technical report, and coordinated submission (Task 4).

2. Task 1: Link Characteristic & Reliability Analysis

2.1 Objective

To evaluate how changes in channel delay and packet error rate affect the throughput of a TCP-based application that operates on a Point-to-Point link.

2.2 Topology

Two nodes were involved in a Point-to-Point connection, which was minimal in nature. The data rate of the link was kept constant at 5 Mbps. Two independent variables, channel delay and receiver-side packet error rate, were considered for the experiment. TCP BulkSendApplication sent data from Node 0 to a PacketSink at Node 1.

2.3 Experimental Procedure

- Step 1: Fix error rate at 0.00 and vary delay across 2 ms, 20 ms, 50 ms, and 100 ms.
- Step 2: Fix delay at 2 ms and vary the packet error rate across 0.00, 0.01, 0.03, and 0.05.
- Step 3: For each run, record Tx bytes, Rx bytes, and throughput from FlowMonitor.

2.4 Results

Delay Setting (ms)	Throughput (Mbps)	Observation
2	4.97	Near full link capacity
20	4.89	Slight throughput decreases; TCP performance remains efficient at low RTT.
50	3.70	Increased RTT begins to noticeably reduce throughput due to slower ACK feedback and congestion window growth.
100	4.38	The throughput is greater than at 50 ms, which goes against the anticipated monotonic decay. It is an artifact due to the length of the simulation, and not a true improvement in performance.

Table 2.1: Latency Observation

Error Rate	Expected Throughput (Mbps)	Measured Throughput (Mbps)	Observation
0.00	5.00	4.98	Near full link capacity
0.01	5.00	4.93	Slight drop, TCP able to handle the 1% loss
0.03	5.00	1.57	Moderate drop, frequent retransmission
0.05	5.00	0.87	Significant degradation, TCP congestion control is very aggressive

Table 2.2: Throughput Under Different Error Rates

2.5 Analysis

There are two ways of degradation. Firstly, the increase in the channel delay leads to a decrease in throughput since TCP throughput is limited by throughput ≈ cWnd / RTT, meaning that the larger RTT makes the process of cWnd growth longer and decreases its speed. Thus, our measurements confirm this assumption since throughput decreased from 4.97 Mbps with 2 ms delay to 3.70 Mbps with 50 ms delay, showing an approximate 25% decline. The 100 ms result (4.38 Mbps) is anomalous because it is higher than the 50 ms result (3.70 Mbps), which contradicts the expected monotonic decline. This is not a real performance gain but an artefact of the short 10-second simulation window. At 100 ms one-way delay (200 ms RTT), the connection only completes roughly 45 RTTs before the simulation ends, and the flow spends a disproportionate fraction of that time in TCP slow-start, where cWnd grows exponentially. Because there is zero packet loss in this run, the flow never enters congestion avoidance and never experiences the linear AIMD growth that would normally dominate steady-state throughput. The average therefore overweights the aggressive slow-start ramp and underweights the steady-state regime that a longer simulation would reveal. A rerun with simTime = 60 s would be expected to show throughput at 100 ms falling below the 50 ms result, consistent with the throughput ≈ cWnd / RTT relationship. The 50 ms result is the more reliable indicator of the true delay-throughput trend in this experiment.

Secondly, even a small fraction of lost packets leads to a dramatic, non-linear collapse in throughput since TCP interprets each loss as congestion and halves cWnd in response. Accordingly, our measurements showed that 1% loss affects throughput only slightly (4.93 Mbps), while 3% loss reduces throughput to 1.57 Mbps (roughly one-third of link capacity) and 5% loss reduces it to 0.87 Mbps (only 17% of the 5 Mbps link capacity). This non-linear collapse corresponds to the Mathis model, where throughput is proportional to $1/\sqrt{p}$.

2.6 Key Takeaway

On a Point-to-Point link, TCP is doubly disadvantaged by delay (which lengthens the feedback loop and slows cWnd growth) and by packet loss (which the protocol misreads as congestion and responds to by halving cWnd). For TechCampus Solutions' WAN links, packet loss is the more dangerous of the two: our measurements show throughput falling to 17% of capacity at just 5% loss, whereas a 25× increase in delay (from 2 ms to 50 ms) only reduces throughput by about 25%. Investigating physical-layer errors on the affected links should therefore be the first priority.

3. Task 2: Shared-Medium Contention Study

3.1 Objective

To observe and document how throughput per node and aggregate Packet Delivery Ratio (PDR) degrade as shared-medium contention on a CSMA LAN increases.

3.2 Topology

One CSMA channel was set up with a speed of 100 Mbps and a 6.56 µs delay between ten nodes. Node 0 was the sink for the UDP, while nodes 1 to 9 were the possible senders. The number of senders was changed to be 1, 3, and 5. All the senders produced constant traffic with 5 Mbps and 1,024-byte UDP packets using OnOffApplication.

3.3 Per-Flow Throughput and PDR

Flow	Source → Sink	Tx Pkts	Rx Pkts	Throughput (Kbps)	PDR (%)
1	10.1.1.2 → 10.1.1.1	4,882	4,882	5,137.71	100.00

Table 3.1: One Sender Active

Flow	Source → Sink	Tx Pkts	Rx Pkts	Throughput (Kbps)	PDR (%)
1	10.1.1.2 → 10.1.1.1	4,882	4,882	5,137.71	100.00
2	10.1.1.3 → 10.1.1.1	4,882	4,882	5,137.65	100.00
3	10.1.1.4 → 10.1.1.1	4,882	4,881	5,136.53	99.98

Table 3.2: Three Senders Active

Flow	Source → Sink	Tx Pkts	Rx Pkts	Throughput (Kbps)	PDR (%)
1	10.1.1.2 → 10.1.1.1	4,882	4,882	5,137.71	100.00
2	10.1.1.3 → 10.1.1.1	4,882	4,882	5,137.36	100.00
3	10.1.1.4 → 10.1.1.1	4,882	4,881	5,136.58	99.98
4	10.1.1.5 → 10.1.1.1	4,882	4,881	5,136.39	99.98
5	10.1.1.6 → 10.1.1.1	4,882	4,882	5,137.53	100.00

Table 3.3: Five Senders Active

Senders	Total Tx	Total Rx	Aggregate Throughput (Kbps)	Per-Node Throughput (Kbps)	Aggregate PDR (%)
1	4,882	4,882	5,137.71	5,137.71	100.0000
3	14,646	14,645	15,411.90	5,137.30	99.9932
5	24,410	24,408	25,685.60	5,137.11	99.9918

Table 3.4: Aggregate Summary

3.4 NetAnim Visualisation

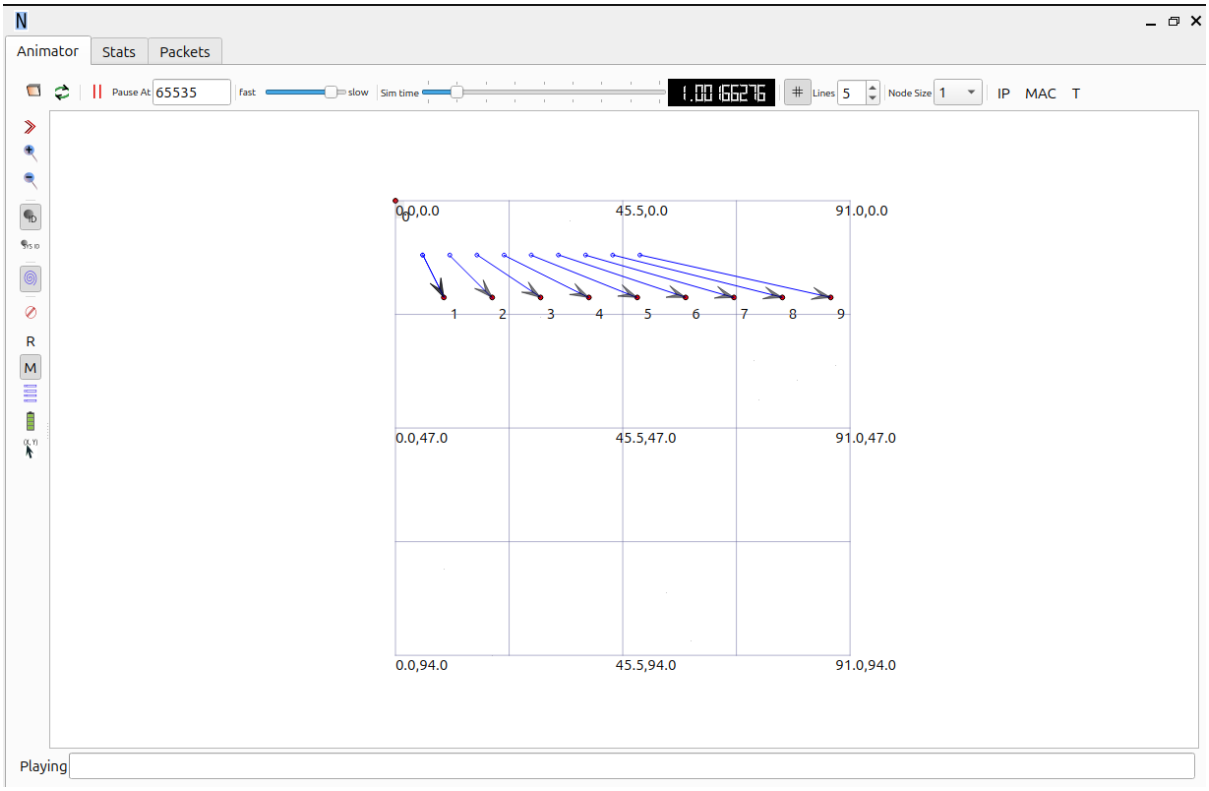


Figure 1: 1 Sender Visualisation

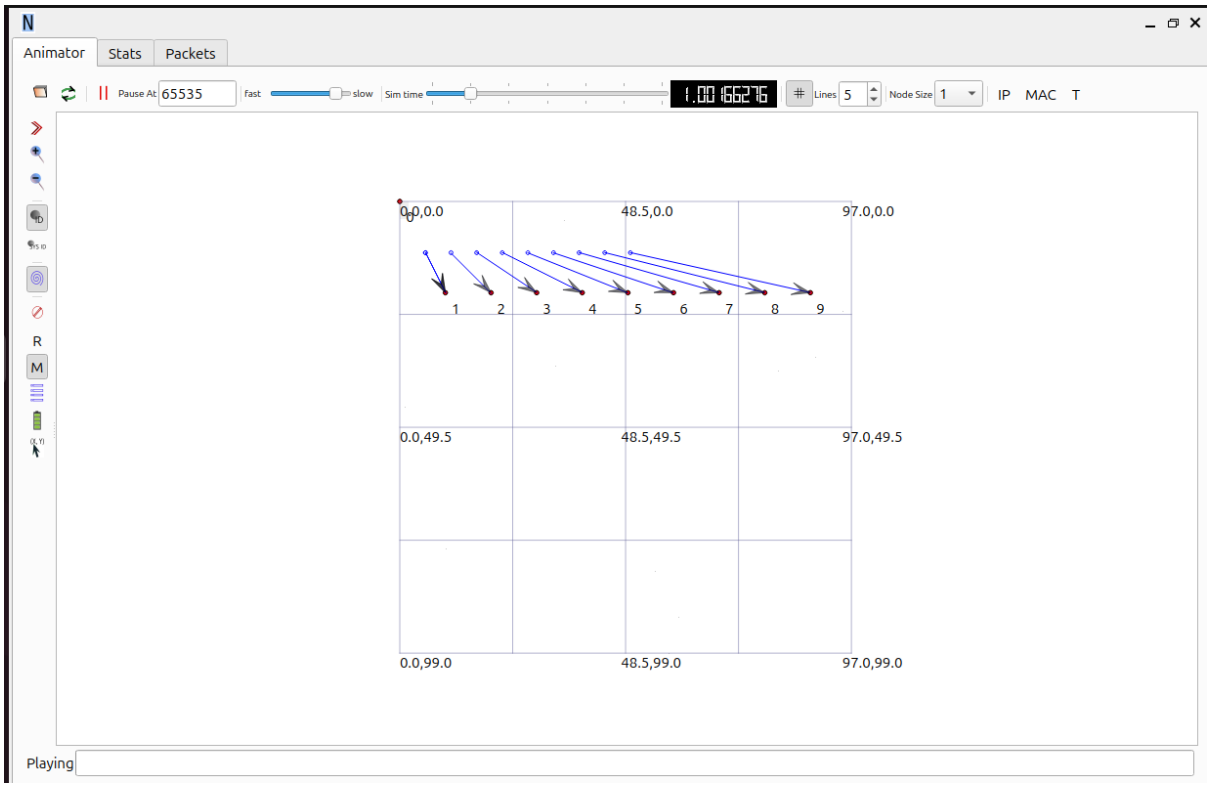


Figure 2: 3 Sender Visualisation

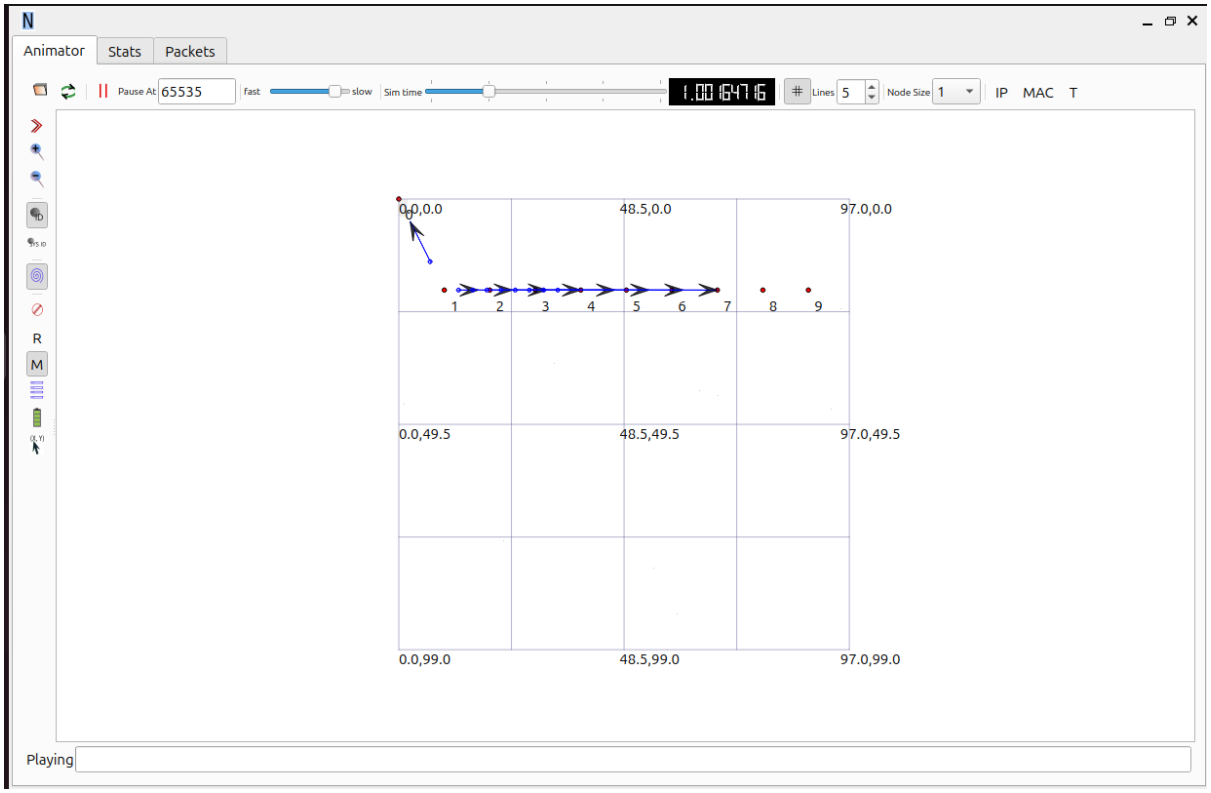


Figure 3: 5 Sender Visualisation

3.5 Analysis

The first observation is that the throughput per node is constant at about 5,137 Kbps regardless of the number of active senders. This is due to the fact that the offered load per sender is relatively low compared to the amount of bandwidth available per sender on a 100 Mbps link with even five senders contending for it. The second observation is that the total throughput increases in a linear fashion from 5.1 Mbps to 15.4 Mbps and finally to 25.7 Mbps. The reason is that the CSMA channel is not yet saturated and therefore contention occurs only on a marginal basis. The third observation is that PDR is perfect in the case of one sender at 100% and falls by 0.0082% when the number of senders is five. This is due to the CSMA/CD collision-resolution back-off where two contenders sense the medium idle simultaneously and thus collide, leading to the dropping of some packets through binary exponential back-off.

In the case where the offered load per sender was increased such that aggregate load exceeds the channel capacity of 100 Mbps, throughput will saturate, and per-sender throughput will fall while PDR will be severely degraded.

3.6 Key Takeaway

CSMA works well below saturation but shows losses due to collisions, which scale in a non-linear fashion once the total offered load starts approaching the channel capacity. For the case of TechCampus Solutions' office LAN, the problem does not seem to be in the medium itself, as it is not currently the bottleneck.

4. Task 3: Transport-Layer Congestion Control

4.1 Objective

To examine two versions of TCP congestion control algorithms, NewReno and CUBIC, that compete for a bottleneck bandwidth of 1 Mbps using graphs generated from their exported congestion window (cWnd).

4.2 Topology

Dumbbell topology was created by using PointToPointDumbbellHelper, with two sending nodes on the left side, two receiving nodes on the right side, and one 1 Mbps bandwidth bottleneck link between the two routers, with 20 ms of propagation delay. The links for leafs were 10 Mbps bandwidth with 2 ms of propagation delay. The first sender node was configured to use TcpNewReno, while the second sender was configured to use TcpCubic.

4.3 Quantitative Comparison

Metric	TCP NewReno	TCP CUBIC	Winner
Peak cWnd before major drop (bytes)	58,000	65,000	CUBIC
cWnd value immediately after drop	0	0	Tied
Time to recover to half of previous peak (s)	Never recovered within simulation window	1.2 s after each drop	CUBIC
Time of first major drop (s)	3.5	7.5	-
Recovery growth pattern	Slow linear (AIMD)	Concave (cubic curve)	CUBIC
Steady-state operating range (bytes)	10,000 – 20,000	28,000 – 35,000	CUBIC
Approximate average cWnd (bytes)	15,000	28,000	CUBIC
Observed behaviour after first drop	Stays suppressed	Recovers and dominates	CUBIC

Table 4.1: cWnd Drop & Recovery Behaviour

4.4 cWnd Curve Analysis

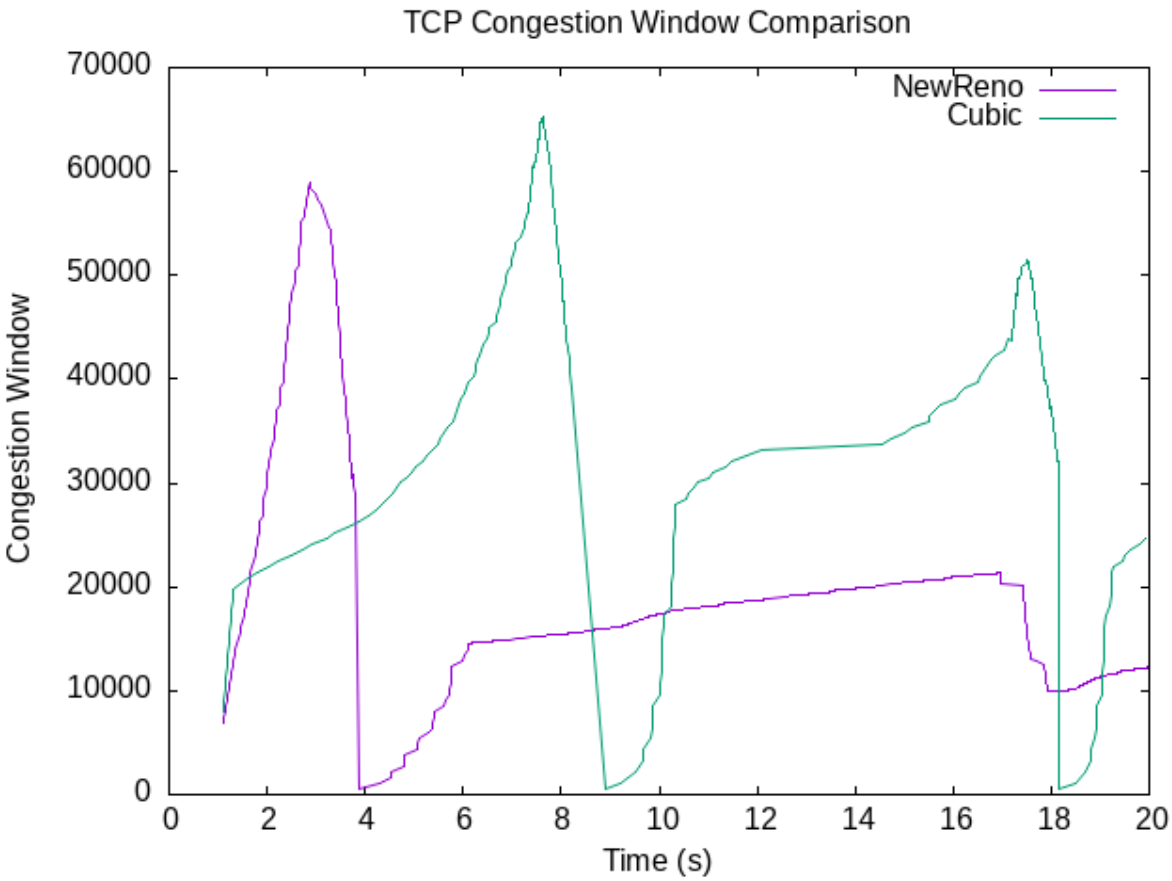


Figure 4: Congestion Window of NewReno vs CUBIC over Time

Figure 4 illustrates the congestion window of both senders during the 20 second simulation period. Three complete congestion cycles are evident. Each cycle consists of one sender's window increasing rapidly, reaching a maximum, then collapsing nearly to zero, suggesting an RTO-based recovery, not fast-retransmit.

The aggressiveness of CUBIC (cyan) is immediately obvious. It reaches a maximum of about 65,000 bytes around t=7.5s, which is significantly larger than NewReno's maximum of about 58,000 bytes reached around t=3.5s. Following each collapse, the recovery of CUBIC exhibits the characteristic concave curve – rapid growth that levels off as it approaches the maximum achieved previously – and ends up with a steady operating region between about 28,000 and 35,000 bytes.

NewReno, on the other hand, after peaking and crashing at t=3.5s, does not recover to that level again. Instead, it operates in a slow-growing, low-bandwidth regime of 10,000-20,000 bytes throughout the remainder of the simulation, growing linearly at one MSS per RTT (additive increase). This graph illustrates the fairness issue that arises when CUBIC and NewReno compete on the same bottleneck link: CUBIC's window increases faster and dominates the link bandwidth, leaving nothing for NewReno.

4.5 Analysis

The CUBIC flow is clearly the more aggressive of the two based on all metrics observed. CUBIC's peak window is roughly 12% larger than NewReno's, but more importantly, its average operating window is approximately double that of NewReno.

The recovery phase reveals more than the peak values do. CUBIC's concave growth curve allows it to return to roughly half of its previous peak within about 1.2 seconds after each drop, whereas NewReno's additive-increase recovery of one MSS per RTT fails to reach even half of its initial 58,000-byte peak for the remainder of the simulation. This recovery asymmetry is the structural reason CUBIC dominates the shared bottleneck.

More telling than the peaks themselves is what happens after each loss event. NewReno cannot reclaim its initial peak because the bottleneck buffer remains saturated by the more aggressive CUBIC flow, so any packets NewReno transmits are dropped before its window can grow. Each drop is then interpreted as further congestion, halving cWnd again and keeping NewReno permanently suppressed. This is a well-known unfairness pattern when CUBIC and NewReno share a bottleneck link.

The fact that both flows collapse to zero rather than to half of their previous values indicates that the losses are recovered via retransmission timeout (RTO) rather than fast retransmit triggered by triple duplicate ACKs.

4.6 Key Takeaway

The selection of the TCP variant on a common bottleneck becomes a crucial factor in determining both performance and fairness. The increased aggressiveness of CUBIC results in over twice the bandwidth shares for CUBIC compared to NewReno on the same 1 Mbps bottleneck. The coexistence of NewReno and CUBIC traffic, as is common in mixed-OS networks, leads to a significant disparity in bandwidth that cannot be addressed at the transport layer using NewReno.

5. Task 4: Synthesis and Cross-Layer Discussion

5.1 Layered View of Degradation

By overlaying all three experiments, we get a composite map of the network breakdown. Each map represents a distinct cause of failure and the failures cited by TechCampus Solutions can be linked to one map.

Layer	Source of Degradation	Observed Symptom (Tasks 1–3)	TechCampus Mapping
Physical / Link	Channel delay & bit-error rate (Task 1)	Throughput falls from 4.97 to 0.87 Mbps under 5% loss. Around 25% reduction between 2 ms and 50 ms delay.	Delayed responses and dropped packets on WAN links
MAC / Data Link	CSMA/CD collisions under contention (Task 2)	PDR falls from 100% to 99.99% as senders rise from 1 to 5	Marginal LAN losses, with a larger issue expected at higher load
Transport	TCP congestion control choice (Task 3)	CUBIC's average cWnd is roughly 2× NewReno's (around 28k vs 15k bytes). NewReno is effectively starved after the first drop.	Unfair bandwidth allocation on bottleneck to remote servers

Table 5.1: Layered View of Network Degradation

5.2 Interactions Between Layers

These three layers are not independent of each other. A high error rate link (from Task 1) is seen by TCP as congestion (from Task 3). So, the transport layer reduces cWnd by half even if there is no congestion at all. Thus, bit-level noise becomes an issue of throughput degradation. In the same way, MAC-layer collisions (as found in Task 2) lead to packet losses that are also perceived as congestion. The result is that even a network with a relatively low level of noise at the physical layer, and collisions at the MAC layer will have much worse performance than the sum of these two effects.

5.3 Targeted Recommendations for TechCampus Solutions

- WAN links (Task 1 problem): examine noise in the physical layer on these links; a packet loss ratio of only 1% cuts the throughput by half. Think about implementing Forward Error Correction at the datalink level to hide temporary errors from TCP.
- Office LAN (Task 2 problem): the current situation with contention fits comfortably into the gentle region of CSMA operation. As soon as the number of users exceeds around 20 simultaneous high-volume senders, switch to a switched architecture to get rid of collisions.

- Bottleneck link to remote servers (Task 3 problem): adopt a consistent TCP version for all your machines (CUBIC, the default Linux one since 2006) to avoid the fairness problems between NewReno and CUBIC highlighted in Task 3. Add Active Queue Management to the bottleneck router (CoDel or PIE).

5.4 Limitations

The RateErrorModel is lossy but ignores bursty losses, CSMA is unaware of the wireless medium, and the dumbbell topology does not consider cross-traffic due to background traffic. Therefore, it can be concluded that the outcome of this research should be regarded as indicative rather than predictive.

6. Conclusion

The project addressed the research question in three interlocking steps. First, Task 1 illustrated how the physical layer parameters, namely the channel delay and packet error rate, have a disproportionately large impact on TCP throughput, such that single-digit percentages of losses lead to throughput reductions by an order of magnitude. Second, Task 2 illustrated how a 100 Mbps CSMA LAN scales gracefully up to five senders at 5 Mbps each, with throughput increasing almost linearly and PDRs staying above 99.99%. Third, Task 3 illustrated that the selection of TCP protocol had a substantial effect on the throughput winners in the case of a bottleneck link, with CUBIC achieving roughly twice the average congestion window of NewReno on a 1 Mbps link.

All three findings have one thing in common, in that network performance is determined by three interrelated variables, which include link quality, MAC contention, and aggressiveness of the transport layer. It is not possible to optimize any of the layers independently. The solution suggested in Section 5.3 addresses the best cure for each symptom identified by TechCampus Solutions.

This project also helps to bring forward an educational principle where the use of discrete event simulation in NS-3 is an effective way to isolate the impact of each layer as it offers full reproducibility, full instrumentation, and variation of one parameter at a time that is not possible in actual network scenarios. In future research, it will be worthwhile to carry out a dumbbell experiment using different TCP protocols like BBR and DCTCP along with active queue management.